

CO4 - AT2 - Coding Golf Assessment

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COURSE: CSA0303 - DATA STRUCTURES

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QUESTIONS
Q1
Bubble Sort
25 marks

Q2
Insertion Sorting
25 marks
Q3
Quick Sorting
25 marks
Q4
Merge Sort
25 marks
1/4 answered

Q1 Bubble Sort 25 marks
Write a C code for performing Bubble Sort - n=5 A[] = {8 2 3 5 7}.
Your Code

```
1 #include <stdio.h>
2 void bubbleSort(int arr[], int n){
3     int i,j,temp;
4     for(i=0;i<n-1;i++){
5         for(j=0;j<n-i-1;j++){
6             if(arr[j]>arr[j+1]){
7                 temp=arr[j];
8                 arr[j]=arr[j+1];
9                 arr[j+1]=temp;
10            }
11        }
12    }
13 }
14 void printArray(int arr[],int n){
15     for(int i=0;i<n;i++){
16         printf("%d",arr[i]);
17         printf("\n");
18     }
19 int main(){
20     int A[] = {8,2,3,5,7};
21     int n=5;
22     printf("Original Array:");
23     printArray(A,n);
24     bubbleSort(A,n);
25     printf("Sorted Array:");
26     printArray(A,n);
```

Input
8 2 3 5 7
Output
Original Array:82357
Sorted Array:23578



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Insertion Sorting
25 marks

Q3
Quick Sorting
25 marks

Q4
Merger Sort
25 marks

1/4 answered

Q2 Insertion Sorting

25 marks

Write a C code for performing Insertion Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int A[5]={8,2,3,5,7};
4     int n=5,i,j,key;
5     for(i=1;i<n;i++){
6         key=A[i];
7         j=i-1;
8         while(j>=0 && A[j]>key){
9             A[j+1] = A[j];
10            j=j-1;
11        }
12        A[j+1]=key;
13    }
14    for(i=0;i<n;i++)
15        printf("%d",A[i]);
16    return 0;
17 }
18
```

Input
8 2 3 5 7

Output
23578

Run

Save

Submit



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QUESTIONS

Q1
Bubbled Sorting
25 marks

Q2
Insertion Sorting
25 marks

Q3
Quick Sorting
25 marks

Q4
Merger Sort
25 marks

3/4 answered

Q3 Quick Sorting

25 marks

Write a C code for performing Quick Sort - n=7 A[] = {8, 2, 3, 5, 7,1,6}.

Your Code

```
1 #include <stdio.h>
2 void quickSort(int A[], int low,int high){
3     if(low>=high) return;
4     int pivot=A[high];
5     int i=low-1;
6     int j,temp;
7     for(j=low;j<high;j++){
8         if(A[j]<=pivot){
9             i++;
10            temp=A[i];
11            A[i]=A[j];
12            A[j]=temp;
13        }
14    }
15    temp=A[i+1];
16    A[i+1]=A[high];
17    A[high]=temp;
18    int pi=i+1;
19    quickSort(A,low,pi-1);
20    quickSort(A,pi+1,high);
21 }
22 int main() {
23     int A[] = {8,2,3,5,7,1,6};
24     int n=7;
25     int i;
26     quickSort(A,0,n-1);
```

Input
8 2 3 5 7 16

Output
Sorted Array:235781616

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QUESTIONS

Q1
Bubble Sort
25 marks**Q2**
Insertion Sort
25 marks**Q3**
Quick Sort
25 marks**Q4**
Merge Sort
25 marks

4/4 answered

Q4 Merge Sort

25 marks

Write a C code for performing Merge Sort - n=8 A[] = {8 2 3 5 7 6 4 1}.

Your Code

```
1 #include <stdio.h>
2
3 void merge(int A[], int l, int m, int r) {
4     int n1 = m - l + 1;
5     int n2 = r - m;
6     int L[n1], R[n2];
7     int i, j, k;
8
9     for (i = 0; i < n1; i++)
10         L[i] = A[l + i];
11     for (j = 0; j < n2; j++)
12         R[j] = A[m + 1 + j];
13
14     i = 0; j = 0; k = l;
15     while (i < n1 && j < n2) {
16         if (L[i] <= R[j]) {
17             A[k] = L[i];
18             i++;
19         } else {
20             A[k] = R[j];
21             j++;
22         }
23         k++;
24     }
25     while (i < n1) {
26         A[k] = L[i];
```

Input

stdin input

Output